

EE 66  
Fall 2026

Signals, Dynamics, and Information

Homework 3

**This homework is due Friday, September 18, at 23:59.**

**Submission Format**

Your homework submission should consist of a single PDF file that contains all of your answers (any hand-written answers should be scanned).

**1. Some (Ortho)Normal Basis Basics**

The set defined by the span of a set of vectors is always a valid subspace for a given vector space. This means that spans are also valid vector spaces. For each of the following vector spaces:

(i) Give the dimension of the space.

(ii) Provide an *orthonormal* basis for the vector space.

$$(a) \mathbb{O} = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} -3 \\ -2 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} \right\}$$

$$(b) \mathbb{P} = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} -4 \\ -4 \\ 0 \end{bmatrix} \right\}$$

$$(c) \mathbb{S} = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \right\}$$

$$(d) \mathbb{T} = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ -3 \end{bmatrix} \right\}$$

## 2. Span Proofs

- (a) Given some set of vectors  $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ , show the following:

$$\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\} = \text{span}\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_1 - \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$$

In other words, we can simultaneously replace  $\mathbf{v}_1$  and  $\mathbf{v}_2$  with their sum and difference without changing the span.

To show this, you have to prove the two following statements:

- If a vector  $\mathbf{q}$  belongs in  $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ , then it must also belong in  $\text{span}\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_1 - \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ .
- If a vector  $\mathbf{r}$  belongs in  $\text{span}\{\mathbf{v}_1 + \mathbf{v}_2, \mathbf{v}_1 - \mathbf{v}_2, \mathbf{v}_3, \dots, \mathbf{v}_n\}$ , then it must also belong in  $\text{span}\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ .

In summary, you have to prove the problem statement from both directions.

*Hint: Can you write  $\mathbf{v}_1$  and  $\mathbf{v}_2$  each as a linear combination of  $(\mathbf{v}_1 + \mathbf{v}_2)$  and  $(\mathbf{v}_1 - \mathbf{v}_2)$ ?*

- (b) Consider the span of the set  $\{\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{u}\}$ . Suppose  $\mathbf{u}$  is in the span of  $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ . Then, show that any vector  $\mathbf{r}$  in  $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n, \mathbf{u}\}$  is in  $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ .

*Hint: Start by applying the definition of span for the  $\mathbf{u}$  and  $\mathbf{r}$  vectors.*

- (c) Let  $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$  be a linearly independent set. Define a new set of vectors by pairing up neighbors and wrapping around:

$$\mathbf{u}_i = \mathbf{v}_i + \mathbf{v}_{i+1} \text{ for } i = 1, \dots, n-1, \quad \mathbf{u}_n = \mathbf{v}_n + \mathbf{v}_1.$$

- (i) Suppose  $n$  is **odd**. Prove that  $\text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\} = \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ .

- (ii) Suppose  $n$  is **even**. Prove that  $\text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\} \neq \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ .

*Hint: If every  $\mathbf{v}$  is a linear combination of all the  $\mathbf{u}_i$ 's, what can you say about  $\text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$  and  $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$ ? What about if every  $\mathbf{u}$  is a linear combination of the  $\mathbf{v}_i$ 's?*

*Hint: Try to solve for  $\mathbf{v}_1$  in terms of the  $\mathbf{u}_i$ 's by substitution: use  $\mathbf{u}_1 = \mathbf{v}_1 + \mathbf{v}_2$  to write  $\mathbf{v}_2$  in terms of  $\mathbf{v}_1$ , then use  $\mathbf{u}_2$  to write  $\mathbf{v}_3$  in terms of  $\mathbf{v}_1$ , and so on around the cycle. You should notice a pattern. Extrapolate to write an equation for, say,  $\mathbf{v}_1$  in terms of all the  $\mathbf{u}_i$ 's. Can you write a similar equation for any  $\mathbf{v}_k$ ? What do you notice about this equation? Does it change depending on the parity of  $n$ ?*

### 3. Signal Spaces: Bounded and Finite-Energy Signals

Consider a complex-valued DT signal  $x : \mathbb{Z} \rightarrow \mathbb{C}$ . We can define  $\ell_1$ ,  $\ell_2$ , and  $\ell_\infty$  norms for these signals:

$$\|x\|_1 = \sum_{n=-\infty}^{\infty} |x[n]|, \quad \|x\|_2 = \left( \sum_{n=-\infty}^{\infty} |x[n]|^2 \right)^{1/2}, \quad \|x\|_\infty = \sup_{n \in \mathbb{Z}} |x[n]|.$$

where  $|\cdot|$  is the complex magnitude:  $|\alpha + i\beta| = \sqrt{\alpha^2 + \beta^2}$  and  $\sup_{n \in \mathbb{Z}}$  denotes the least upper bound on the signal (this is often functionally the same as  $\max_{n \in \mathbb{Z}}$ ).

As an engineer you'll encounter lots of signals, and you'll probably ask the following questions about them: *Will it ever blow up?* and *Will it go on forever?* Consider the feedback whine of a microphone, a resounding "yes!" to both questions! To help answer these mathematically, we can evaluate the following conditions:

- **Absolutely summable:**  $\ell_1(\mathbb{Z}) = \{x : \|x\|_1 < \infty\}$ . The magnitudes  $|x[n]|$  add up to a finite number, so  $x[n]$  must tend to 0 as  $|n| \rightarrow \infty$ . This is the strongest condition.
- **Finite energy:**  $\ell_2(\mathbb{Z}) = \{x : \|x\|_2 < \infty\}$ . The squared magnitudes  $|x[n]|^2$  add up to a finite number, so  $x[n]$  must also tend to 0 as  $|n| \rightarrow \infty$ . This is a weaker condition, but perhaps more physical.
- **Bounded:**  $\ell_\infty(\mathbb{Z}) = \{x : \|x\|_\infty < \infty\}$ . For some finite  $M$  and all  $n$ ,  $|x[n]| \leq M$ . Here,  $x[n]$  doesn't need to tend to 0. This is the weakest condition, but it is often the most commonly used!

#### Useful facts:

- $\sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$  for  $|r| < 1$ .
  - $\sum_{k=1}^{\infty} \frac{1}{k^s}$  is finite if and only if  $s > 1$ . In particular,  $\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}$ .
  - $u[n]$  is the unit step:  $u[n] = 1$  for  $n \geq 0$ , and 0 otherwise.
- (a) For each of the following signals, find  $\|x\|_1$ ,  $\|x\|_2$ , and  $\|x\|_\infty$  (write  $\infty$  if a norm diverges), and thus determine the strictest condition it satisfies. Also, verify that  $\|x\|_\infty \leq \|x\|_2 \leq \|x\|_1$ .
- $x[n] = \left(-\frac{1}{2}\right)^n u[n]$
  - $x[n] = \frac{1}{n+1} u[n]$
  - $x[n] = e^{i\omega_0 n}$  for a fixed real  $\omega_0$
  - $x[n] = n u[n]$
- (b) Let  $y[n] = x[-n]$  (time reversal) and  $z[n] = x[n-k]$  (a shift by an integer  $k$ ). Show that  $\|y\|_p = \|z\|_p = \|x\|_p$  for  $p = 1, 2, \infty$ .
- (c) Show that the set of bounded signals,  $\ell_\infty(\mathbb{Z}) = \{x : \|x\|_\infty < \infty\}$ , is a subspace of the set of all DT signals. Consider also the set of *unbounded* signals,  $\mathcal{U} = \{x : \|x\|_\infty = \infty\}$ . Is *this* a subspace? Why or why not?

*Hint: Just as vectors add and scale entry by entry, DT signals add and scale sample by sample. It might be helpful to bound  $|x[n] + y[n]|$  by a constant that does not depend on  $n$ .*

*Hint: to show a set is not a subspace, it is enough to exhibit one subspace axiom that fails.*

#### 4. Periodicity of Complex Exponentials

A discrete-time signal  $x : \mathbb{Z} \rightarrow \mathbb{C}$  is  $p$ -periodic if it repeats every  $p$  timesteps. That is,

$$x[n+p] = x[n] \quad \forall n \in \mathbb{Z}. \quad (1)$$

Note that Equation 1 holds for all multiples of periods (like  $p, 2p, 3p, \dots$ ). The smallest  $p \in \mathbb{Z}^+$  for which Equation 1 holds is called the fundamental period of  $x[n]$ : the shortest length after which the signal's pattern first repeats.

Let  $a : \mathbb{Z} \rightarrow \mathbb{R}$  be a discrete-time signal with fundamental period  $p_a$  and  $b : \mathbb{Z} \rightarrow \mathbb{R}$  be a discrete-time signal with fundamental period  $p_b$ , where  $p_a$  and  $p_b$  are coprime positive integers (that is,  $p_a$  and  $p_b$  have no common factors other than 1).

- (a) Let  $c[n] = 100a[n]$ . Show that the fundamental period of  $c$  is  $p_a$ .
- (b) Let  $d[n] = e^{ip_a n}$ . Is  $d$  periodic? If so, what is its fundamental period?
- (c) Now let  $g[n] = e^{2\pi i n}$ . Is  $g$  periodic? If so, what is its fundamental period?
- (d) Let

$$f[n] = \frac{1}{2} \exp\left(\frac{2\pi i}{p_a} n\right).$$

Is  $f$  periodic? If so, what is its fundamental period?

- (e) Let  $h[n] = f[n] + b[n]$ . Show that  $h$  is  $(p_a p_b)$ -periodic.
- (f) Now instead suppose that  $p_a$  is an integer multiple of  $p_b$ , so  $p_a = k p_b$  for some  $k \in \mathbb{Z}^+$ . Is  $h$  still periodic? Give a period of  $h$ .
- (g) Based on the previous two questions, what can you say about a period of the sum of two periodic signals? Be sure to address the role of coprimality.
- (h) Now consider

$$q[n] = \cos^2\left(\frac{2\pi}{p_a} n\right).$$

Assume that  $p_a$  is even. Rewrite  $q$  using complex exponentials and the inverse Euler formula, and show that  $\frac{p_a}{2}$  is a period of  $q$ .

- (i) Finally, consider

$$y[n] = \cos\left(\frac{2\pi}{p_a} n\right) \cos\left(\frac{2\pi}{p_b} n\right),$$

where  $p_a$  and  $p_b$  are coprime. Rewrite  $y$  using complex exponentials and show that it is  $(p_a p_b)$ -periodic. What does this example suggest about a period of the product of two periodic signals?

## 5. Homework Reflection

Doing homework is an **investment in your education**, and the course staff wants to hear about your experience with this problem set! After finishing the rest of the homework, fill out these brief reflections on what you got out of your time. We'll read this and take your answers into account as we write future assignments, in this semester and future ones.

Note that we are *genuinely curious* to read what you have to write. We expect *brief but thoughtful* responses. **Hence, this part will be weighed significantly in your homework completion grade.**

- (a) Did you work on this homework alone or in a group? Did this work well for you? *We encourage you to attempt each problem on your own first, and to work with a group when you're stuck.*
- (b) What resources did you use to solve the homework problems? How did you use these resources, and how did they help you? Feel free to mention LLM tools, if you've used any.
- (c) What was the most fun / interesting problem on this homework? What was the least fun / un-interesting problem?
- (d) Did you feel prepared to tackle these problems? Were there any problems that seemed completely disjoint from what you've seen in class? Was there anything covered in class that you weren't able to practice in this assignment?
- (e) Is there anything else you would like to share?